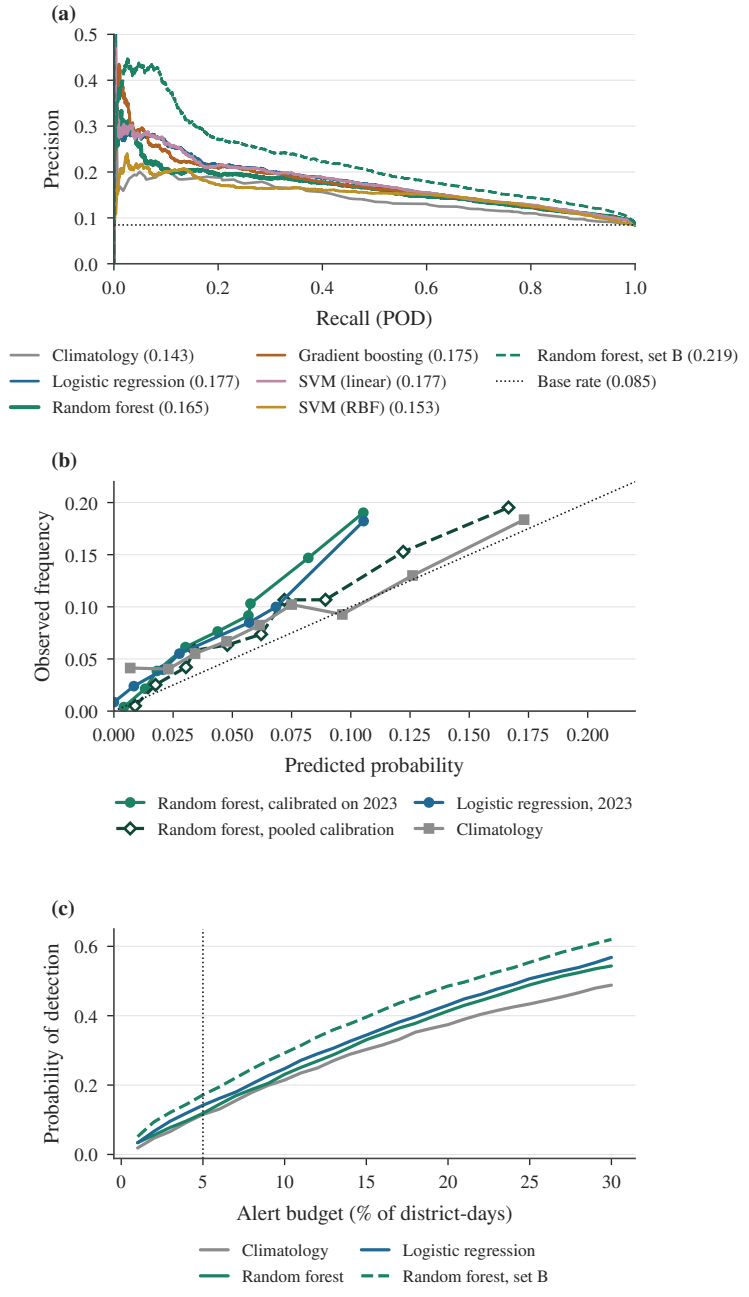

(a)
Precision
0.0
0.1
0.2
0.3
0.4
0.5
0.0
0.2
0.4
0.6
0.8
1.0
Recall (POD)
Climatology (0.143)
Logistic regression (0.177)
Random forest (0.165)
Gradient boosting (0.175)
SVM (linear) (0.177)
SVM (RBF) (0.153)
Random forest, set B (0.219)
Base rate (0.085)
(b)
Observed frequency
0.00
0.05
0.10
0.15
0.20
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Predicted probability
Random forest, calibrated on 2023
Logistic regression, 2023
Random forest, pooled calibration
Climatology
(c)
Probability of detection
0.0
0.2
0.4
0.6
0
5
10
15
20
25
30
Alert budget (% of district-days)
Climatology
Logistic regression
Random forest
Random forest, set B